

EXPERIMENT 3

CHECKING OUT THE BONDS



AIM

To investigate some metallic, covalent and ionic bonds.

MATERIALS

- 2 petri dishes
- 3 × 100 mL beakers
- Bunsen burner, tripod and gauze mat
- tongs
- 2 stirrers
- 3 crucibles
- water
- some salt, sugar and soft metal solder
- hand lens or stereomicroscope

METHOD

- 1 Pour 50 mL of water into each of the beakers.
- 2 Add salt, sugar and solder to separate beakers.
- 3 Add more of each substance to its beaker and stir until no more dissolves. Record your observations.
- 4 For each of the substances that dissolved, pour

just enough solution into a petri dish to cover the bottom. Place somewhere warm to dry out.

5 Set up your Bunsen burner and tripod.

6 Place a sample of each substance into separate crucibles.

7 Heat each substance and record what you observe before, during and after heating.

8 Try to crush some of each substance on your bench top. Record your results.

9 Once the liquid in the petri dishes has evaporated, scrape the bottom of each dish and record what you see. You may need a hand lens or stereomicroscope.

DISCUSSION

After reading about metallic, ionic and covalent bonds, which substances do you think have these types of bonds? Give reasons for your answer.

In covalent substances, the atoms don't gain or lose electrons, but they do share electrons with the other atoms they are bonded with. This sharing of electrons gives each atom the same number of electrons as a noble gas and makes it more stable.

The bonds that hold the atoms together in molecules are covalent bonds. Covalent bonds also are found in very hard substances like diamonds.

Ionic bonding

Another important kind of bonding is **ionic bonding**. This takes place between metals and non-metals. In this case the metal becomes more stable by giving away one or more of its outer electrons. This leaves it positively charged. The non-metal becomes more stable by capturing the metal's discarded electron or electrons, making it negatively charged overall because of the extra electrons.

Remember, an atom which has a charge is called an ion.

Now, the two stable ions are strongly attracted to each other because of their positive and negative charges. Large num-

bers of positive and negative ions will form an ionic lattice held together by this strong electrical attraction (Figure 2.8).

Ionic substances tend to form hard crystals which are very rigid and brittle. Salt is one of these, with strong attraction between the metal sodium (Na) ions and non-metal chlorine (Cl) or chloride ions, which is why it is also called sodium chloride (NaCl).

Questions



- 1 Explain the difference between metallic, covalent and ionic bonding. Examples and diagrams should be used where possible.
- 2 Why do metals lose electrons easily and non-metals gain electrons easily?
- 3 Why can't an ionic compound be bent and shaped?
- 4 What are the major differences between molecules and lattices?